	Resistance = [illegible]	

6a	**Magnetic flux density** of a magnetic field is defined as the **force per unit length per unit current** acting on a **straight** current-carrying conductor placed **perpendicular to a magnetic field**.	2
6bi	Since a counter balance (rider) is needed, force on BC is downward. By Fleming's left-hand rule, the magnetic field is towards the right, which would be due to a current in the clockwise direction as viewed from the eye.	1 1
6bii	$B = \mu_0 nI \qquad = \mu_0 \left(\frac{1000}{0.500}\right)(3.0)$ $= 7.5 \times 10^{-3}$ T	1 1
6biii	$F = BIL = 7.54 \times 10^{-3} \times 3.0 \times 0.040$ $= 9.0 \times 10^{-4}$ N	1
6biv	Let the distance of the rider from the axis XY be *x*. Apply Principle of moments about the axis XY,	 1

	$F_{BC}(AB) = mgx$ $x = \dfrac{(9.05\times10^{-4})(0.20)}{(0.40\times10^{-3})(9.81)} = 0.046$ m = 4.6 cm	1

7a		1